



On the characterization of the space of derivations in evolution algebras

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Abstract

We study the space of derivations for some finite-dimensional evolution algebras, depending on the twin partition of an associated directed graph. For evolution algebras with a twin-free associated graph, we prove that the space of derivations is zero. For the remaining families of evolution algebras, we obtain sufficient conditions under which the study of such a space can be simplified. We accomplish this task by identifying the null entries of the respective derivation matrix. Our results suggest how strongly the associated graph's structure impacts in the characterization of derivations for a given evolution algebra. Therefore, our approach constitutes an alternative to the recent developments in the research of this subject. As an illustration of the applicability of our results, we provide some examples and we exhibit the classification of the derivations for non-degenerate irreducible three-dimensional evolution algebras.

Keywords Genetic algebra · Evolution algebra · Derivation · Graph · Twin partition

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1 Introduction

The evolution algebras became an interesting class of non-associative genetic algebras because of their many applications and connections to other areas of mathematics. The first reference of a Theory of Evolution Algebras is due to Tian and Vojtechovsky, see [26], who in 2006 state the first properties for these mathematical structures. Further on, Tian, in his seminal work [24], gave an interesting correspondence between this and other subjects of mathematics like, for example, the Theory of Discrete-time Markov Chains. We refer the reader to [2–11, 14–16, 22, 25] and references therein for an overview of recent results, applications, and interesting open problems.

In order to state the first definitions, let $\Lambda := \{1, \dots, n\}$. An n -dimensional $\mathbb{K}$ -algebra $\mathcal{A}$ is called an *evolution algebra* if it admits a basis $B = \{e_i\}_{i \in \Lambda}$ such that $e_i \cdot e_j = 0$ whenever $i \neq j$. A basis with this property is known as *natural basis*. The scalars $\omega_{ij} \in \mathbb{K}$ such that

$$e_i^2 = e_i \cdot e_i = \sum_{k \in \Lambda} \omega_{ik} e_k$$

are called the *structure constants* of $\mathcal{A}$ relative to B , and the matrix $M_B = (\omega_{ik})$ is called the *structure matrix* of $\mathcal{A}$ relative to B . Our purpose is to study the derivations of an evolution algebra. Given an (evolution) algebra $\mathcal{A}$, a *derivation* of $\mathcal{A}$ is a linear map $d : \mathcal{A} \rightarrow \mathcal{A}$ such that

$$d(u \cdot v) = d(u) \cdot v + u \cdot d(v),$$

for all $u, v \in \mathcal{A}$. The space of all derivations of the (evolution) algebra $\mathcal{A}$ is denoted by $\text{Der}(\mathcal{A})$. In [24] it is observed that a linear map d such that $d(e_i) = \sum_{k=1}^n d_{ik} e_k$ is a derivation of the evolution algebra $\mathcal{A}$ if, and only if, it satisfies the following conditions:

$$\omega_{jk} d_{ij} + \omega_{ik} d_{ji} = 0, \quad \text{for } i, j, k \in \Lambda \text{ such that } i \neq j, \quad (1)$$

$$\sum_{k=1}^n \omega_{ik} d_{kj} = 2\omega_{ij} d_{ii}, \quad \text{for } i, j \in \Lambda, \quad (2)$$

where $M_B = (\omega_{ij})$ is the structure matrix of $\mathcal{A}$ relative to a natural basis B . Therefore, (1) and (2) are the starting point whether one want to obtain a description of the space of derivations of an evolution algebra $\mathcal{A}$. We point out that such a space is a Lie algebra which may be used as a tool for studying the structure of the original algebra. For some genetic algebras, the space of all the derivations has already been described in [12, 13, 17–20, 23]. In the particular case of an evolution algebra, a complete characterization of such space is still an open question. We recommend to the reader the references [1, 6, 8, 9, 15, 21] for a good idea of the existing results about this topic.

In [8] the authors prove that the space of derivations of n -dimensional complex evolution algebras with non-singular matrices is zero, and they describe the space of derivations of evolution algebras with matrices of rank $n - 1$. In [15] the authors extend the description of the derivations of evolution algebras with non-singular matrices to the case of fields with any characteristic. Although the approaches considered in [1, 6, 9] are different, they provide a useful contribution to the field. While [9] gives a complete characterization for

the space of derivation on two-dimensional evolution algebras, and [1, 21] consider the case of three-dimensional solvable and finite-dimensional nilpotent evolution algebras, [6] do it for the case of evolution algebras associated with graphs. We point out that the later includes examples of algebras with singular matrices with any rank so their arguments can be seen as an alternative to the ones developed by [8, 15]. Indeed, the approach stated in [6] relies only on the structural properties of the considered graph. In this work, we extend this approach to study the derivations of a given evolution algebra. We accomplish this task by exploring the fact that, as pointed by [14], any evolution algebra induces a directed graph. Moreover, we illustrate the applicability of our approach by characterizing the derivations of non-degenerate irreducible three-dimensional evolution algebras.

The paper is organized as follows. In Sect. 2 we review some of the standard notation of Evolution Algebras and Graph Theory, we state our main results, and we illustrate their applicability through examples. Section 3 is devoted to the statement of auxiliary results and the proofs of our main theorems.

2 Main results

2.1 Preliminary definitions and notation

In order to present our results, we start with basic definitions and notation for directed graphs appearing in [2]. A *directed graph* is a 4-tuple $E = (E^0, E^1, s_E, r_E)$ where E^0, E^1 are sets and $s_E, r_E : E^1 \rightarrow E^0$ are maps. The elements of E^0 are called the *vertices* of E , and the elements of E^1 the *arrows* or *directed edges* of E . For $f \in E^1$ the vertices $r(f)$ and $s(f)$ are called the *range* and the *source* of f , respectively. If E^0 and E^1 are both finite, we say that E is *finite*. A vertex $v \in E^0$ is called a *sink* if it verifies that $s(f) \neq v$, for every $f \in E^1$. A *path* or a *path from $s(f_1)$ to $r(f_n)$* in E , μ , is a finite sequence of arrows $\mu = f_1 \dots f_n$ such that $r(f_i) = s(f_{i+1})$ for $i \in \{1, \dots, (n-1)\}$. In this case, we say that n is the *length* of the path μ and denote by μ^0 the set of its vertices, i.e., $\mu^0 := \{s(f_1), r(f_1), \dots, r(f_n)\}$. Let $\mu = f_1 \dots f_n$ be a path in E . If $|\mu| = n \geq 1$, and if $v = s(\mu) = r(\mu)$, then μ is called a *closed path based at v* . If $\mu = f_1 \dots f_n$ is a closed path based at v and $s(f_i) \neq s(f_j)$ for every $i \neq j$, then μ is called a *cycle based at v* or simply a *cycle*. A cycle of length 1 will be said to be a *loop*. Given a finite graph E , its *adjacency matrix* is the matrix $Ad_E = (a_{ij}) \in \mathbb{Z}^{(E^0 \times E^0)}$ where a_{ij} is the number of arrows from i to j .

There are several ways to associate a graph with an evolution algebra (see [2, 14]). We consider the directed graph described in [2] as follows. Given a natural basis $B = \{e_i\}_{i \in A}$ of an evolution algebra $\mathcal{A}$ and its structure matrix $M_B = (\omega_{ij}) \in M_A(\mathbb{K})$, consider the matrix $P = (a_{ij}) \in M_A(\mathbb{K})$ such that $a_{ij} = 0$ if $\omega_{ij} = 0$ and $a_{ij} = 1$ if $\omega_{ij} \neq 0$. The *graph associated with the evolution algebra $\mathcal{A}$* (relative to the basis B), denoted by $E_{\mathcal{A}}^B$ (or simply by E if the algebra $\mathcal{A}$ and the basis B are understood), is the directed graph whose adjacency matrix is given by $P = (a_{ij})$.

Example 1 Let $\mathcal{A}$ be an evolution algebra with natural basis $B = \{e_1, e_2, e_3\}$ such that

$$\begin{aligned} e_1^2 &= 2e_1 + e_2, \\ e_2^2 &= -e_1 + 3e_3, \\ e_3^2 &= 3e_3. \end{aligned}$$

The associated directed graph is given in Fig. 1.

The *underlying graph* of the directed graph G is the a graph which is obtained from G after forgetting the orientation of their arrows. A directed graph G is *connected* if their underlying graph is connected. On the other hand, an evolution algebra $\mathcal{A}$ is *reducible* if there exist two nonzero ideals I and J such that $\mathcal{A} = I \oplus J$. In other case, it is called *irreducible*. An evolution algebra $\mathcal{A}$ is *non-degenerate* if there is a natural basis B such that $e_i^2 \neq 0$ for all $e_i \in B$. In general, the connectivity of the graph associated with an evolution algebra depends on the chosen natural basis, as [14, Example 2.5] shows. It is well known by [14, Lemma 2.7] the fact that a given algebra is non-degenerate does not depend on the chosen natural basis. Connectedness is related to irreducibility by [14, Proposition 2.8] which states that a finite non-degenerate evolution algebra $\mathcal{A}$ with a natural bases B is irreducible if, and only if, the associated graph $E_{\mathcal{A}}^B$ is connected.

The following definitions can be found in [2, Definition 3.1], but we include them here for the sake of completeness. Let $B = \{e_i\}_{i \in \Lambda}$ and $M_B = (\omega_{ij})$ be, respectively, a natural basis and the structure matrix relative to B of an evolution algebra $\mathcal{A}$. For any $i_0 \in \Lambda$, the *first-generation descendants* of i_0 is the set given by $D^1(i_0) := \{k \in \Lambda : \omega_{i_0 k} \neq 0\}$. In addition, given a subset $U \subseteq \Lambda$, we let $D^1(U) := \{j \in \Lambda : j \in D^1(i) \text{ for some } i \in U\}$. If $E = (E^0, E^1, r, s)$ is the graph associated with $\mathcal{A}$, observe that

$$D^1(i_0) = \{k \in \Lambda : \exists f \in E^1 \mid s(f) = v_{i_0}, r(f) = v_k\},$$

with $v_{i_0}, v_k \in E^0$.

By analogy with Graph Theory, we define the following notions. Let $\mathcal{A}$ be an evolution algebra with natural basis B and let $i, j \in \Lambda$. We say that i and j are *twins relative to B* if $D^1(i) = D^1(j)$. If $\mathcal{A}$ has no twins relative to B , that is, if

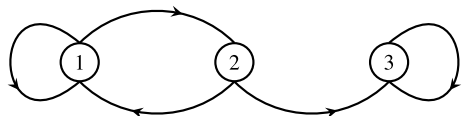
$$\text{for all } i, j \in \Lambda, i \neq j, D^1(i) \neq D^1(j),$$

then we say that $\mathcal{A}$ is *twin-free relative to B* (see Example 1). We notice that by defining the relation $\sim_{t_B}$ on the set of indices Λ by $i \sim_{t_B} j$ whether i and j are twins relative to B , then $\sim_{t_B}$ is an equivalence relation. An equivalence class of the twin relation $\sim_{t_B}$ is referred to as a *twin class relative to B* . In other words, the twin class of an index i is the set $\{j \in \Lambda : i \sim_{t_B} j\}$. The set of all twin classes relative to B of Λ is denoted by $\Pi_B(\Lambda)$, and it is referred to as the *twin partition relative to B* of Λ . These definitions depend on the chosen natural basis as the following example shows.

Example 2 Let $\mathcal{A}$ be an evolution algebra with natural basis $B = \{e_1, e_2, e_3\}$ such that

$$\begin{aligned} e_1^2 &= e_2, \\ e_2^2 &= e_3, \\ e_3^2 &= e_3. \end{aligned}$$

Fig. 1 Associated directed graph of the evolution algebra of Example 1



In this natural basis, we have $D^1(1) = \{2\}$, and $D^1(2) = D^1(3) = \{3\}$. Then, $\Pi_B(A) = \{\{1\}, \{2, 3\}\}$. Now consider a new natural basis given by $B' = \{f_1, f_2, f_3\}$ with $f_1 = e_1$, $f_2 = e_2 - e_3$ and $f_3 = e_2 + e_3$. In the basis B' we have $D^1(1) = D^1(2) = D^1(3) = \{2, 3\}$ so $\Pi_{B'}(A) = \{\{1, 2, 3\}\}$. In Fig. 2 we represent the associated graphs for this evolution algebra.

2.2 Characterization of the space of derivations

We focus on non-degenerate irreducible evolution algebras with finite dimension. In other words, we consider evolution algebras such that their associated graph is finite, connected and without sinks. As mentioned before, our aim is to study the space of derivations. In the rest of the paper, we shall assume that $\mathcal{A}$ is a n -dimensional $\mathbb{K}$ -evolution algebra with $\text{char}(\mathbb{K}) = 0$. We denote the associated graph to $\mathcal{A}$ by $E = (E^0, E^1, r, s)$, so $|E^0| = n$. We suppose that $d \in \text{Der}(\mathcal{A})$ is such that

$$d(e_i) = \sum_{k=1}^n d_{ik} e_k, \quad (3)$$

with $d_{ik} \in \mathbb{K}$ for every $i, k \in A$.

Proposition 1 *Let $\mathcal{A}$ be an n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in A}$ and structure matrix (ω_{ij}) . If $d \in \text{Der}(\mathcal{A})$, then d satisfies the following conditions:*

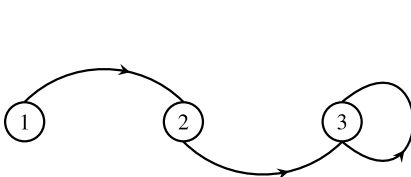
(i) *If $i, j \in A$, $i \neq j$, and $D^1(i) \cap D^1(j) \neq \emptyset$ then*

$$d_{ij} = -\frac{\omega_{ik}}{\omega_{jk}} d_{ji}, \quad \text{for } k \in D^1(i) \cap D^1(j).$$

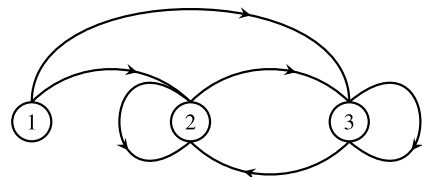
(ii) *If $i, j \in A$, $i \neq j$, and $D^1(i) \cap (D^1(j))^c \neq \emptyset$ then $d_{ji} = 0$.*

(iii) *If $\mathcal{A}$ is non-degenerate, then for every $i, j \in A$, $i \neq j$ with $D^1(i) \cap D^1(j) = \emptyset$ we have $d_{ij} = d_{ji} = 0$.*

(iv) *For any $i \in A$, we have*



(a) Associated directed graph of the evolution algebra relative to B



(b) Associated directed graph of the evolution algebra relative to B'

Fig. 2 Associated directed graphs of the evolution algebra of Example 2 according to the chosen basis

$$\sum_{k \in D^1(i)} \omega_{ik} d_{kj} = \begin{cases} 0, & \text{if } j \notin D^1(i), \\ 2\omega_{ij} d_{ii}, & \text{if } j \in D^1(i). \end{cases}$$

Proof In order to prove (i) consider $i, j \in \Lambda$, with $i \neq j$, such that $D^1(i) \cap D^1(j) \neq \emptyset$, and let $k \in D^1(i) \cap D^1(j)$. Then, $\omega_{ik} \neq 0$ and $\omega_{jk} \neq 0$. Therefore, condition (1) implies

$$d_{ij} = -\frac{\omega_{ik}}{\omega_{jk}} d_{ji}.$$

Analogously, assume now that for some $i, j \in \Lambda$ we have $D^1(i) \cap (D^1(j))^c \neq \emptyset$, and let $k \in D^1(i) \cap (D^1(j))^c$. This means that $\omega_{ik} \neq 0$ while $\omega_{jk} = 0$. This implies, by (1), that $d_{ji} = 0$ and the proof of (ii) is completed. Item (iii) is a consequence of (ii) and $\mathcal{A}$ is non-degenerate since $D^1(i) \cap D^1(j) = \emptyset$ implies $D^1(i) \cap (D^1(j))^c \neq \emptyset$ and $D^1(j) \cap (D^1(i))^c \neq \emptyset$. Finally, item (iv) may be obtained by observing in (2) that for any $i \in \Lambda$, $\omega_{ik} \neq 0$ if and only if, $k \in D^1(i)$. $\square$

Corollary 1 Let $\mathcal{A}$ be an n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$. Let $d \in \text{Der}(\mathcal{A})$. If $d_{ij} = 0$ and $i \sim_{t_B} j$ then $d_{ji} = 0$.

Proof The proof is straightforward from Proposition 1(i). $\square$

Lemma 1 Let $\mathcal{A}$ be a non-degenerate n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Let $d \in \text{Der}(\mathcal{A})$. If $d_{ij} = d_{ji} = 0$ for any $i, j \in \Lambda$, $i \neq j$, then $d_{ii} = 0$ for any $i \in \Lambda$.

Proof Let $i \in \Lambda$ such that $d_{ii} \neq 0$. If $j \in D^1(i)$ applying Proposition 1(iv), we have that

$$2\omega_{ij} d_{ii} = \sum_{k \in D^1(i)} \omega_{ik} d_{kj} = \omega_{ij} d_{jj}.$$

Therefore,

$$2d_{ii} = d_{jj}, \quad \text{for } j \in D^1(i). \quad (4)$$

Now, we will see that a directed graph E that has no sinks contains at least one cycle. Indeed, let $u_1 \in E^0$ an arbitrary vertex. If there exists an arrow $g \in E^1$ such that $s(g) = r(g) = u_1$ then we get the desired result. On the contrary, since u_1 is not a sink, there exist a vertex $u_2 \in E^0 \setminus \{u_1\}$ and arrow f_1 such that $s(f_1) = u_1$ and $r(f_1) = u_2$. Reasoning in the same way with u_2 , we get that there exists either a cycle of length 1 or a cycle of length 2 or there exist a vertex $u_3 \in E^0 \setminus \{u_1, u_2\}$ and an arrow $f_2 \in E^1$ such that $s(f_2) = u_2$ and $r(f_2) = u_3$. Since the algebra has dimension n , we iterate this process at most $(n-1)$ times and either we have a cycle or we have a path $f_1 f_2 \dots f_{n-1}$ that contains n different vertices $\{u_1, \dots, u_n\}$. Since u_n is not a sink, necessarily there exist a vertex $u_i \in E^0$ and an arrow f_n such that $s(f_n) = u_n$ and $r(f_n) = u_i$. Therefore, in any case, we get a cycle.

Let $c_1, \dots, c_s$ be the cycles of E and $c_1^0, \dots, c_s^0$ the set of vertices of such cycles, that is $c_i^0 = \{v_{i_1}, \dots, v_{i_{t_i}}\}$ with $v_{ij} \in E^0$ for every $i \in \{1, \dots, s\}$ and $j \in \{1, \dots, t_i\}$. For $j \in \{1, \dots, (t_i - 1)\}$ and $i \in \{1, \dots, s\}$, we have that $i_{j+1} \in D^1(i_j)$ and $i_1 \in D^1(i_{t_i})$ then by (4)

$$2d_{ij} = d_{i_{j+1}j_{j+1}} \quad \text{and} \quad 2d_{i_j i_j} = d_{i_1 i_1}.$$

Therefore, $d_{ij} = 0$ for all $j \in \{1, \dots, t_i\}$ and $i \in \{1, \dots, s\}$. Suppose that there exists a vertex v_k such that $v_k \in E^0 \setminus (\cup_{i=1}^s c_i^0)$. Since v_k is not a sink, there exist an $i_0 \in \{1, \dots, s\}$ and $h \in \{1, \dots, t_{i_0}\}$ with $v_{i_0 h} \in c_{i_0}^0$ and a path μ with $\mu^0 = \{v_k, v_{k_1}, \dots, v_{k_l}, v_{i_0 h}\}$ such that $v_{k_i} \in E^0 \setminus (\cup_{i=1}^s c_i^0)$ for all $i \in \{1, \dots, l\}$. Since $d_{i_0 h i_0 h} = 0$, then $d_{k_i k_i} = 0$ for all $i \in \{1, \dots, l\}$ and therefore $d_{kk} = 0$. $\square$

Lemma 2 Let $\mathcal{A}$ be a non-degenerate n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Let $d \in \text{Der}(\mathcal{A})$. If $d_{ij} \neq 0$ for $i, j \in \Lambda$, $i \neq j$ then $i \sim_{t_B} j$.

Proof Let $i, j \in V$, $i \neq j$ such that $d_{ij} \neq 0$. By Proposition 1 (iii), we have that $D^1(i) \cap D^1(j) \neq \emptyset$ then by Proposition 1 (i) we have that

$$d_{ji} = -\frac{\omega_{jk}}{\omega_{ik}} d_{ij} \neq 0 \quad \text{for } k \in D^1(i) \cap D^1(j).$$

Then, by Proposition 1 (ii),

$$D^1(i) \cap (D^1(j))^c = \emptyset \quad \text{and} \quad D^1(j) \cap (D^1(i))^c = \emptyset.$$

Therefore, $D^1(i) = D^1(j)$. $\square$

Theorem 1 Let $\mathcal{A}$ be an irreducible non-degenerate n -dimensional evolution algebra. Let $B = \{e_i\}_{i \in \Lambda}$ be a natural basis of $\mathcal{A}$ with structure matrix $M_B = (\omega_{ij})$, and let $d \in \text{Der}(\mathcal{A})$. If $\mathcal{A}$ is twin-free relative to B , then $d = 0$.

Proof Let $d \in \text{Der}(\mathcal{A})$. Since $\mathcal{A}$ is twin-free relative to B , we have that $i \sim_{t_B} j$ for any $i, j \in \Lambda$, with $i \neq j$. This in turn implies, by Lemma 2, $d_{ij} = d_{ji} = 0$, for any $i, j \in \Lambda$, with $i \neq j$. Therefore, by Lemma 1 we conclude $d = 0$. $\square$

The previous theorem gives us a first taste of how strongly the associated graph's structure impacts in the existence, or not, of nonzero derivations for a given evolution algebra. The theorem gains in interest if we realize that it provides a simple criterion for the identification of evolution algebras for which the only derivation is the null map. Moreover, it provides an alternative to previous analysis related in the literature about this topic. For the sake of comparison, we illustrate the applicability of Theorem 1 with some examples.

Example 3 Consider the evolution algebra $\mathcal{A}$ with natural basis $B = \{e_1, e_2, e_3, e_4, e_5\}$ such that

$$\begin{aligned} e_1^2 &= e_1 + 2e_2 + e_4 - e_5, \\ e_2^2 &= -3e_1 + 3e_3 + 2e_5, \\ e_3^2 &= e_2 + e_4 - 3e_5, \\ e_4^2 &= -2e_3 + e_1 + e_4 + e_5, \\ e_5^2 &= -e_1 + e_2 + 2e_3 - 5e_4, \end{aligned}$$

It is not difficult to see that the associated graph (see Fig. 3a) is twin-free relative to B . In this case, $D^1(1) = \{1, 2, 4, 5\}$, $D^1(2) = \{1, 3, 5\}$, $D^1(3) = \{2, 4, 5\}$, $D^1(4) = \{1, 3, 4, 5\}$ and $D^1(5) = \{1, 2, 3, 4\}$. Therefore, Theorem 1 implies that if $d \in \text{Der}(\mathcal{A})$ then $d = 0$. Notice that the same result may be obtained as a consequence of [8, Theorem 2.1] or [15, Theorem 4.1(1)] since the structure matrix is a non-singular matrix.

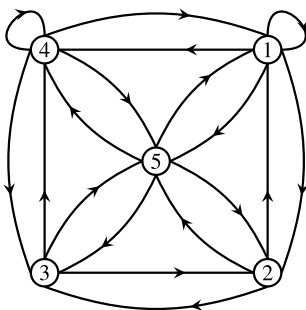
Example 4 Consider the evolution algebra $\mathcal{A}$ with natural basis $B = \{e_1, e_2, e_3, e_4\}$ such that

$$\begin{aligned} e_1^2 &= e_1 + e_2, \\ e_2^2 &= e_2 + e_4, \\ e_3^2 &= e_1 + 2e_2 + e_4, \\ e_4^2 &= e_1 - e_4. \end{aligned}$$

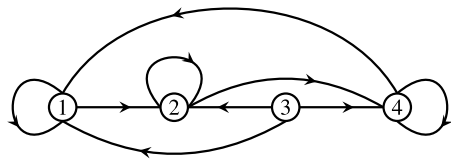
Note that $\mathcal{A}$ is twin-free relative to B since $D^1(1) = \{1, 2\}$, $D^1(2) = \{2, 4\}$, $D^1(3) = \{1, 2, 4\}$ and $D^1(4) = \{1, 4\}$. Therefore, Theorem 1 implies that $d = 0$ for all $d \in \text{Der}(\mathcal{A})$. Contrary to what happens in Example 3, we cannot apply results from [8, 15] because this case is an example of an n -dimensional evolution algebra whose structure matrix has rank $n - 2$; which is a case not covered by those works. Observe that since $\mathcal{A}$ is a four-dimensional evolution algebra, it is not nilpotent, and since there exists $\omega_{ij} \notin \{0, 1\}$, we cannot apply the results from [1, 6, 9, 21] neither.

We emphasize that, given an evolution algebra, the twin partition of its associated graph plays an important role in the description of its space of derivations. Our approach extends the one developed by [6] for the particular class of evolution algebras associated with graphs. Our next task is to explore derivations of those evolution algebras with at least one twin class with more than one vertex.

Theorem 2 Let $\mathcal{A}$ be a non-degenerate n -dimensional evolution algebra. Let $B = \{e_i\}_{i \in \Lambda}$ be a natural basis of $\mathcal{A}$ with structure matrix $M_B = (\omega_{ij})_{i,j \in \Lambda}$, and let $\Pi_B(\Lambda) = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_k\}$ be the twin partition of Λ relative to B , with $k \in \{1, \dots, n\}$. If $d \in \text{Der}(\mathcal{A})$, then



(a) Associated graph for the evolution algebra of Example 3



(b) Associated graph for the evolution algebra of Example 4.

Fig. 3 Associated twin-free graphs of evolution algebras

$$d = A_1 \oplus A_2 \oplus \cdots \oplus A_k,$$

where A_i is a $|\mathcal{T}_i|$ -dimensional square matrix, for $i \in \{1, \dots, k\}$.

Proof It is enough to show that $d_{ij} = 0$ provided i, j belong to different twin classes. This is a consequence of Lemma 2 because by taking $i, j \in \Lambda$, $i \neq j$, such that $i \sim_B j$ then $d_{ij} = d_{ji} = 0$. Thus, one can identify each matrix with a twin class after a suitable ordering of the generators index set. $\square$

The principal significance of Theorem 2 is in the assertion that the derivations of an evolution algebra are strongly related to the twin partition of its associated directed graph. It claims that the induced matrix of a derivation is the direct sum of matrices induced by the twin classes of the respective graph. This is one step in the direction of a generalized version of Theorem 2.6 in [6]. However, it is worth pointing out that, while in [6] the twin classes of at most two vertices induce only null matrices, here we could have a nonzero matrix associated with such twin classes.

Example 5 Let $\mathcal{A}$ be an evolution algebra with natural basis $B = \{e_1, e_2, e_3\}$ such that

$$\begin{aligned} e_1^2 &= e_3, \\ e_2^2 &= -e_3, \\ e_3^2 &= e_1 + e_2. \end{aligned}$$

In this natural basis, we have that $D^1(1) = D^1(2) = \{3\}$ and $D^1(3) = \{1, 2\}$. Then, $\Pi_B(\Lambda) = \{\mathcal{T}_1, \mathcal{T}_2\}$, where $\mathcal{T}_1 = \{1, 2\}$, $\mathcal{T}_2 = \{3\}$, and it is not difficult to see that

$$d = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

belongs to $\text{Der}(\mathcal{A})$. Therefore, this is an example of evolution algebra with twin classes of size at most two and such that $d = A_1 \oplus A_2$ with $A_1 \neq 0$ and $A_2 \neq 0$.

From now on, we shall investigate the behavior of the matrices appearing in Theorem 2.

2.3 The null entries of a derivation matrix when restricted to a twin class

As we shall see later, to know that some elements in the matrix representation of a given derivation are equal to zero allow us to have some information about the remaining elements. Let us start with some general conditions for a given pair $i, j \in \Lambda$ to guarantee $d_{ij} = 0$. Then, we restrict our attention to those elements inside a given twin class.

Proposition 2 Let $\mathcal{A}$ be a n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Let $d \in \text{Der}(\mathcal{A})$ and consider $i, j \in \Lambda$, with $i \neq j$. Assume that $|D^1(i)| \geq 2$ and that there exist $k, \ell \in D^1(i)$ such that

$$\det \begin{pmatrix} \omega_{ik} & \omega_{jk} \\ \omega_{i\ell} & \omega_{j\ell} \end{pmatrix} \neq 0.$$

Then, $d_{ij} = d_{ji} = 0$.

Proof Assume $|D^1(i)| \geq 2$ and note that $k, \ell \in D^1(i)$ implies by Proposition 1(i) that

$$d_{ij} = -\frac{\omega_{ik}}{\omega_{jk}}d_{ji}, \quad \text{and} \quad d_{ij} = -\frac{\omega_{i\ell}}{\omega_{j\ell}}d_{ji}.$$

This in turns implies

$$(\omega_{ik}\omega_{j\ell} - \omega_{i\ell}\omega_{jk})d_{ji} = 0,$$

so by hypothesis we obtain $d_{ji} = 0$ and again by Proposition 1(i) we have $d_{ij} = 0$. $\square$

Proposition 3 Let $\mathcal{A}$ be a n -dimensional evolution algebra with $n \geq 3$. Let $B = \{e_i\}_{i \in \Lambda}$ be a natural basis of $\mathcal{A}$ with structure matrix $M_B = (\omega_{ij})$, and let $d \in \text{Der}(\mathcal{A})$. Consider $i, j \in \Lambda$, with $i \neq j$. If

$$\left(\sum_{k \in \Lambda} \omega_{ik}^2\right)\left(\sum_{k \in \Lambda} \omega_{jk}^2\right) - \left(\sum_{k \in \Lambda} \omega_{ik}\omega_{jk}\right)^2 \neq 0, \quad (5)$$

then $d_{ij} = d_{ji} = 0$.

Proof Fix $i, j \in \Lambda$, with $i \neq j$. If we consider the homogeneous system $\omega_{jk}d_{ij} + \omega_{ik}d_{ji} = 0$, for $k \in \Lambda$, with variables d_{ij} , then the system is an overdetermined system. By applying the least-squares solution, we get the homogeneous system

$$(d_{ji}d_{ij}) \begin{pmatrix} \sum_{k \in \Lambda} \omega_{ik}^2 & \sum_{k \in \Lambda} \omega_{ik}\omega_{jk} \\ \sum_{k \in \Lambda} \omega_{ik}\omega_{jk} & \sum_{k \in \Lambda} \omega_{jk}^2 \end{pmatrix} = 0$$

which has trivial solution provided its determinant is nonzero. $\square$

Remark 1 Proposition 3 holds for evolution algebras of dimension three or higher. In the case of a two-dimensional evolution algebra, the above proposition means as follows if the evolution algebra is perfect, i.e., $A = A^2$, then $d_{12} = d_{21} = 0$ being the natural basis $B = \{e_1, e_2\}$.

Example 6 Let $\mathcal{A}$ be an evolution algebra and a natural basis B with product

$$\begin{aligned} e_1^2 &= (1/2)e_1 - (1/4)e_2, \\ e_2^2 &= -2e_1 + e_2, \\ e_3^2 &= 2e_1 + e_2. \end{aligned}$$

Note that we have a unique twin class, that is, $\Pi_B(A) = \{\mathcal{T}_1\}$, where $\mathcal{T}_1 = \{1, 2, 3\}$. The proposition above implies that $d_{13} = d_{31} = d_{23} = d_{32} = 0$, which simplifies the task of calculating the derivations. Indeed, a straightforward calculation shows that the derivations are given by

$$\begin{pmatrix} d_{11} & -(1/2)d_{11} & 0 \\ -2d_{11} & d_{11} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Notice that in the previous example we have a unique twin class. Moreover, observe that in this example $i \in D^1(i)$ whether $i \in \{1, 2\}$, and we can write $d = A_1 = B_1 \oplus B_2$ where the B_i matrices are related to those vertices with and without loops in the associated directed graph, respectively. This suggests that a good strategy to understand the behavior of the matrices A_i in Theorem 2 could be focusing in the existence of loops in each directed graph associated with each twin partition.

Proposition 4 *Let $\mathcal{A}$ be an n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$. Let $\mathcal{T}$ be a twin class relative to B of Λ such that $\mathcal{T} = \{i\}$, for certain $i \in \Lambda$. If $i \in D^1(i)$ then $d_{ii} = 0$.*

Proof This is a direct consequence of Proposition 1(iv) and Lemma 2. Indeed, if we take $j = i$ we obtain, as $i \in D^1(i)$, the following expression:

$$\omega_{ii}d_{ii} + \sum_{k \in D^1(i) \cap \mathcal{T}} \omega_{ik}d_{ki} = 2\omega_{ii}d_{ii},$$

but, Lemma 2 implies $d_{ki} = 0$ for any $k \in \mathcal{T}$. Therefore, $d_{ii} = 0$.

From now on, our approach is to subdivide a twin class into two disjoint sets formed by those vertices with a loop (we use wl in the notation) and those vertices without a loop (we use nl in our notation). $\square$

Proposition 5 *Let $\mathcal{A}$ be an n -dimensional evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$. Let $\mathcal{T}$ be a twin class relative to B of Λ . Assume that $\mathcal{T} = \mathcal{T}^{wl} \cup \mathcal{T}^{nl}$, where*

$$\mathcal{T}^{wl} := \{i \in \mathcal{T} : i \in D^1(i)\} \neq \emptyset,$$

and $\mathcal{T}^{nl} := \mathcal{T} \setminus \mathcal{T}^{wl}$. Also, assume that $d_{ij} = 0$ for any $i, j \in \mathcal{T}^{nl}$, $i \neq j$. Then,

- (i) $d_{ii} = 0$ for any $i \in \mathcal{T}$.
- (ii) if $j \in \mathcal{T}^{nl}$ and $i \in \mathcal{T}^{wl}$ we have

$$\sum_{k \in \mathcal{T}^{nl}} w_{ik}d_{kj} = 0.$$

In other words, if $W^{(wl)} := (w_{ij})_{i,j \in \mathcal{T}^{nl}}$ and $\tilde{D} := (d_{ij})_{i \in \mathcal{T}^{wl}, j \in \mathcal{T}^{nl}}$ then $W^{(wl)}\tilde{D} = 0$.

- (iii) *If $W^{(wl)} = (w_{ij})_{i,j \in \mathcal{T}^{wl}}$ is a non-singular matrix, then $d_{ij} = 0$ for any i, j such that $i \in \mathcal{T}^{wl}$ or $j \in \mathcal{T}^{nl}$.*

Proof Let us prove (i). We consider two separate cases.

Let $i \in \mathcal{T}^{nl}$. Then, Proposition 1 (iv) (by assuming $j = i$) implies that

$$2w_{ii}d_{ii} = \sum_{k \in D^1(i)} w_{ik}d_{ki} = \sum_{k \in D^1(i) \cap \mathcal{T}} w_{ik}d_{ki} + \sum_{k \in D^1(i) \cap \mathcal{T}^{nl}} w_{ik}d_{ki}.$$

But

$$\sum_{k \in D^1(i) \cap \mathcal{T}} w_{ik}d_{ki} = \sum_{k \in \mathcal{T}^{nl}} w_{ik}d_{ki},$$

while

$$\sum_{k \in D^1(i) \cap \mathcal{T}} w_{ik} d_{ki} = 0,$$

because $k \in D^1(i) \cap \mathcal{T}$ and i belong to different twin classes, so $d_{ki} = 0$. Then, we conclude that

$$2w_{ii}d_{ii} = \sum_{k \in \mathcal{T}^l} w_{ik}d_{ki} = w_{ii}d_{ii},$$

because we are assuming $d_{ki} = 0$ whether $k, i \in \mathcal{T}^{nl}, k \neq i$. Then, $d_{ii} = 0$.

Let $i \in \mathcal{T}^l$. Proposition 1 (iv) (by assuming $j \in \mathcal{T}^{nl}$) implies that

$$2w_{ij}d_{ii} = \sum_{k \in D^1(i)} w_{ik}d_{kj} = \sum_{k \in D^1(i) \cap \mathcal{T}} w_{ik}d_{kj} + \sum_{k \in D^1(i) \cap \mathcal{T}^l} w_{ik}d_{kj}.$$

In this case, we obtain reasoning as before

$$2w_{ij}d_{ii} = \sum_{k \in \mathcal{T}^{nl}} w_{ik}d_{kj} = w_{ij}d_{jj} + \sum_{k \in \mathcal{T}^l, k \neq j} w_{ik}d_{kj} = w_{ij}d_{jj} = 0,$$

where $d_{jj} = 0$ because of the previous case. So $d_{ii} = 0$.

Now let us consider (ii). Take $i \in \mathcal{T}^{nl}$ and $j \in \mathcal{T}^l$. We apply Proposition 1 (iv) to obtain

$$\sum_{k \in D^1(i)} w_{ik}d_{kj} = 2w_{ij}d_{ii} = 0.$$

Then,

$$0 = \sum_{k \in D^1(i)} w_{ik}d_{kj} = \sum_{k \in D^1(i) \cap \mathcal{T}} w_{ik}d_{kj} + \sum_{k \in D^1(i) \cap \mathcal{T}^l} w_{ik}d_{kj} = \sum_{k \in \mathcal{T}^{nl}} w_{ik}d_{kj},$$

and the proof is complete. We note that condition (ii) may be written in matricial form. The result is the statement of condition (iii). $\square$

Example 7 Let $\mathcal{A}$ an evolution algebra with product

$$\begin{aligned} e_1^2 &= e_1 - e_2 + e_3, \\ e_2^2 &= e_1 + e_2 - e_3, \\ e_3^2 &= e_1 + e_2 - e_3, \\ e_4^2 &= -e_1 - e_2 + e_3, \end{aligned}$$

and let $d \in \text{Der}(\mathcal{A})$. We point out that the structure matrix has rank 2 so it is not covered by [8, 15]. Note that this case is not covered by [1, 5, 9, 21] neither. Here, we have $\Pi_B(A) = \{\mathcal{T}\}$, where $\mathcal{T} = \{1, 2, 3, 4\}$. Moreover, $\mathcal{T}^{nl} = \{1, 2, 3\}$ and $\mathcal{T}^l = \{4\}$, and we can apply Proposition 3 to conclude that $d_{14} = d_{41} = 0$ and $d_{ij} = d_{ji} = 0$ provided $i, j \in \mathcal{T}^{nl}, i \neq j$. Then, Proposition 5(i) implies $d_{ii} = 0$ for any $i \in \mathcal{T}$. Finally, Proposition 5(ii) and (1) implies that d is equal to

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & d_{24} \\ 0 & 0 & 0 & d_{24} \\ 0 & d_{24} & d_{24} & 0 \end{pmatrix}.$$

Example 8 Consider the evolution algebra $\mathcal{A}$ with natural basis $B = \{e_1, e_2, e_3, e_4, e_5\}$ such that

$$\begin{aligned} e_1^2 &= e_1 + 2e_2 + 3e_3, \\ e_2^2 &= e_1 + e_2 + 3e_3, \\ e_3^2 &= 2e_1 + e_2 + e_3, \\ e_4^2 &= -2e_1 + e_2 - e_3, \\ e_5^2 &= e_1 + e_2 + e_3. \end{aligned}$$

This is another example of n -dimensional evolution algebra whose structure matrix has rank equals to $n - 2$. Therefore, the results in [8, 15] do not apply, and the results in [6, 9] neither. Here, we have $\Pi_B(\Lambda) = \{\mathcal{T}\}$ with $\mathcal{T}^{nl} = \{1, 2, 3\}$ and $\mathcal{T}^l = \{4, 5\}$. By Proposition 3 we have $d_{ij} = 0$ provided $i, j \in \mathcal{T}^{nl}$, $i \neq j$. In addition, by Proposition 5(iii) we conclude that $d_{ij} = 0$ provided i or j belongs to $\mathcal{T}^{nl}$. In fact, note that

$$W^{(wl)} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{pmatrix},$$

is a non-singular matrix. Finally, Proposition 2 allows to conclude that $d_{ij} = 0$ for $i, j \in \mathcal{T}^l$ and therefore if $d \in \text{Der}(\mathcal{A})$ then $d = 0$.

Lemma 3 Let $\mathcal{A}$ be a non-degenerate n -evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Suppose that there exists a twin class $\mathcal{T}$ such that $\mathcal{T}^{nl} = \emptyset$. Let $i \in \mathcal{T}$ and suppose that for every $j_1, j_2 \in D^1(i)$, $\mathcal{T}_{j_1} \neq \mathcal{T}_{j_2}$. Moreover, we assume that $d_{jj} = 0$ for certain $j \in D^1(i)$. Then, $d_{ii} = 0$ for every $i \in \mathcal{T}$.

Proof Note that $d_{j_1 j_2} = d_{j_2 j_1} = 0$ for every $j_1, j_2 \in D^1(i)$ because $j_1 \sim_{t_B} j_2$. On the other hand, by Proposition 1(iv) we have that

$$\sum_{k \in D^1(i)} \omega_{ik} d_{kj} = 2\omega_{ij} d_{ii}$$

which implies that $\omega_{ij} d_{jj} = 2\omega_{ij} d_{ii}$ since $d_{kj} = 0$ if $k \neq j$. Finally, as $d_{jj} = 0$ we get that $d_{ii} = 0$ for every $i \in \mathcal{T}$. $\square$

Lemma 4 Let $\mathcal{A}$ be an n -evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Suppose that there exist a twin class $\mathcal{T}$ and an element $i \in \mathcal{T}$ such that $D^1(j) = \{i\}$ for every $j \in \mathcal{T}$. Then, $d_{ji} = d_{ji} = d_{ii} = d_{jj} = 0$ for every $j \in \mathcal{T}$.

Proof Let $j \in \mathcal{T} \setminus \{i\}$. Note that $j \in \mathcal{T}^l$ and $i \in \mathcal{T}^{nl}$. By Proposition 5, we get $\omega_{ii} d_{ij} = 0$ therefore $d_{ij} = 0$. By Corollary 1, $d_{ji} = 0$. Moreover, by (2), $\omega_{ii} d_{ii} = 2\omega_{ii} d_{ii}$ so $d_{ii} = 0$. Again, by (2), $\omega_{ji} d_{ii} = 2\omega_{ji} d_{jj}$ then $d_{jj} = 0$. $\square$

Lemma 5 Let $\mathcal{A}$ be an n -evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Let $\mathcal{T}$ be a twin class relative to B . Suppose that there exists $k \in \Lambda$ such that $D^1(k) \cap \mathcal{T} = \{j\}$ for certain $j \in \mathcal{T}$. Then, $d_{jl} = d_{lj} = 0$ for every $l \in \mathcal{T} \setminus \{j\}$.

Proof Let $\mathcal{T}$ be a twin class and $k \in \Lambda$ which verifies $D^1(k) \cap \mathcal{T} = \{j\}$. By (2), $\omega_{kj}d_{jl} = 2\omega_{kl}d_{kk}$ for every $l \in \mathcal{T}$ because $d_{jl} = 0$ for every $l \in \mathcal{T}$ (see Lemma 2) and $D^1(k) \cap \mathcal{T} = \{j\}$. If $l \in \mathcal{T} \setminus \{j\}$, then $\omega_{kj}d_{jl} = 0$. Therefore, $d_{jl} = 0$. Moreover, by Lemma 1 we get $d_{lj} = 0$. $\square$

Lemma 6 Let $\mathcal{A}$ be an evolution algebra with a natural basis $B = \{e_i\}_{i \in \Lambda}$ and structure matrix $M_B = (\omega_{ij})$. Let $\mathcal{T}$ be a twin class relative to B , and we consider $k \in D^1(\mathcal{T})$. Suppose that $D^1(k) \subseteq \mathcal{T} \cup \{k\}$. Moreover, we suppose that $d_{sk} = d_{ks} = 0$ for every $s \in D^1(\mathcal{T}) \setminus \{k\}$. If $D^1(\mathcal{T}) \cap \mathcal{T} = \{i, j\}$ for certain $i, j \in \mathcal{T}$ with $i, j \neq k$, then $d_{ji} = d_{ij} = d_{ii} = d_{jj} = d_{kk} = 0$.

Proof First, we will see that $d_{kk} = 0$. Note that $k \in \mathcal{T}$. By (2) and $D^1(k) \subseteq \mathcal{T} \cup \{k\}$, we have that

$$\sum_{h \in \mathcal{T}} \omega_{kh}d_{hk} + \omega_{kk}d_{kk} = 2\omega_{kk}d_{kk}.$$

Applying Lemma 2, we get that $d_{kk} = 0$. If we consider the descendants of i and (2), we get that

$$\sum_{h \in D^1(\mathcal{T}) \setminus \{k\}} \omega_{ih}d_{hk} + \omega_{ik}d_{kk} = 2\omega_{ik}d_{ii}.$$

Since $d_{kk} = 0$ this implies that $d_{ii} = 0$. Similarly, $d_{jj} = 0$. On the other hand, again by (2)

$$\omega_{ii}d_{ii} + \omega_{ij}d_{ji} + \sum_{h \in \mathcal{T}} \omega_{ih}d_{hi} = 2\omega_{ii}d_{ii}.$$

But $d_{hi} = 0$ for every $h \in \mathcal{T}$ applying Lemma 2 and $d_{ii} = 0$ therefore $d_{ji} = 0$ and by Lemma 1 $d_{ij} = 0$. $\square$

3 Derivations of non-degenerate irreducible three-dimensional evolution algebras

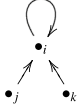
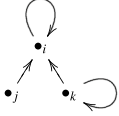
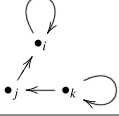
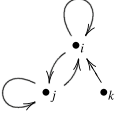
In this section, we apply our results to identify the derivations of any non-degenerate and irreducible three-dimensional evolution algebra. In order to accomplish this task, we consider such an evolution algebra, and we consider its associated directed graph. We know from [14] that any evolution algebra with a given natural basis induces a unique directed graph. Moreover, our results show how strong is the connection between the structure of such associated graph and the derivations of the considered evolution algebra. As we state in Theorem 1 if the graph is twin-free, then the only derivation is the null map. Thus, we will focus our attention in studying those cases where the associated graph is not twin-free. Since we have only 3 vertices, the only possibilities are that we have a twin class with three or with two elements.

The table below shows the associated graph and the derivations of the corresponding three-dimensional evolution algebras depending on the number of the nonzero entries in its structure matrix i.e., the number of arrows in the associated graph. For each case, we identify the elements, and the descendants, in the biggest twin class.

Section 2.3 summarizes many sufficient conditions to guarantee the existence of null elements in the derivation matrix of a given evolution algebra. In what follows, we shall see that through a suitable application of these results we can identify many cases of evolution algebras which the only derivation is the null map.

Indeed, those evolution algebras of type 2, 4, 5 and 6 have derivations zero as a consequence of Lemmas 4, 2, 1 and 1, in that order. On the other hand, the evolution algebras associated with the types 17 and 21 have derivations zero by Lemma 6. Also, notice that any evolution algebra of type 3, 8 to 12, 14, 15, 16, 18, 20 or 22 has derivations equal to zero by Lemmas 5, 2, 1 and 1, in that order. On the other hand, the first column, the first row and the main diagonal of the derivation matrix associated with the evolution algebras of type 1 are zero because of Lemma 4. Finally, the zeros in the derivation matrix associated with the evolution algebras of type 7 and 19 are consequence of Lemma 2.

We point out that after the application of our results the task of identifying derivations is reduced to the study of only 5 from the 23 different evolution algebras whose directed graph is not twin-free.

Type	Number of arrows	Twin class: $\mathcal{T}$	$D^1(\mathcal{T})$	Graph	Derivation
1	3	$\{i, j, k\}$	$\{i\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & d_{jk} \\ 0 & -\frac{\omega_{ki}}{\omega_{ji}} d_{jk} & 0 \end{pmatrix}$
2	3	$\{i, j\}$	$\{i\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
3	3	$\{i, j\}$	$\{k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
4	4	$\{i, j\}$	$\{i\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
5	4	$\{i, j\}$	$\{i\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
6	4	$\{i, j\}$	$\{i\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
7	4	$\{i, j\}$	$\{k\}$		$\begin{pmatrix} d_{ii} & \frac{3\omega_{kj}}{\omega_{ki}} d_{ii} & 0 \\ \frac{3\omega_{ki}}{\omega_{kj}} d_{ii} & d_{ii} & 0 \\ 0 & 0 & 2d_{ii} \end{pmatrix}$
8	4	$\{i, j\}$	$\{k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
9	5	$\{i, j\}$	$\{i, j\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Type	Number of arrows	Twin class: $\mathcal{T}$	$D^1(\mathcal{T})$	Graph	Derivation
10	5	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
11	5	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
12	5	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
13	6	$\{i, j, k\}$	$\{i, j\}$		$\begin{pmatrix} d_{ii} & \frac{d_{ii}\omega_{ij}}{\omega_{ii}} & d_{ik} \\ \frac{d_{ii}\omega_{ji}}{\omega_{ji}} & d_{ii} & -\frac{d_{ik}\omega_{ji}}{\omega_{ii}} \\ \frac{\omega_{ij}}{\omega_{ii}\omega_{ki}} & \frac{d_{ik}\omega_{ki}}{\omega_{jj}} & d_{ii} \end{pmatrix}$
14	6	$\{i, j\}$	$\{i, j\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
15	6	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
16	6	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
17	7	$\{i, j\}$	$\{i, j, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
18	7	$\{i, j\}$	$\{i, j, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Type	Number of arrows	Twin class: $\mathcal{T}$	$D^1(\mathcal{T})$	Graph	Derivation
19	7	$\{i, j\}$	$\{i, j\}$		$\begin{pmatrix} -\frac{\omega_{jj}d_{ij}}{\omega_{ii}} & d_{ij} & 0 \\ \frac{\omega_{ji}d_{ij}}{\omega_{ii}} & \frac{\omega_{ji}d_{ij}}{\omega_{ii}} & 0 \\ 0 & 0 & 0 \end{pmatrix}$
20	7	$\{i, j\}$	$\{i, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
21	8	$\{i, j\}$	$\{i, j, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
22	8	$\{i, j\}$	$\{i, j, k\}$		$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
23	9	$\{i, j, k\}$	$\{i, j, k\}$		See *

$$* \begin{pmatrix} -\left(\frac{\omega_{jj}d_{ij}}{\omega_{ii}} + \frac{\omega_{kk}d_{ik}}{\omega_{ii}}\right) & d_{ij} & d_{ik} \\ -\frac{\omega_{jj}d_{ij}}{\omega_{ij}} & -\left(\frac{\omega_{jj}d_{ij}}{\omega_{ii}} + \frac{\omega_{kk}d_{ik}}{\omega_{ii}}\right) & -\left(\frac{\omega_{jk}d_{ij}}{\omega_{ii}} + \frac{(\omega_{jk}\omega_{kk} + \omega_{ji}\omega_{ii})d_{ik}}{\omega_{ii}\omega_{jj}}\right) \\ -\frac{\omega_{kk}d_{ik}}{\omega_{ik}} & \frac{-\omega_{jj}}{\omega_{jk}}\left(\frac{\omega_{jj}}{\omega_{ii}} + \frac{\omega_{ji}}{\omega_{jj}}\right)d_{ij} - \frac{\omega_{kk}d_{ik}}{\omega_{ii}} & -\left(\frac{\omega_{jj}d_{ij}}{\omega_{ii}} + \frac{\omega_{kk}d_{ik}}{\omega_{ii}}\right) \end{pmatrix}$$

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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